

Here is a step-by-step guide to Hosting a Static Website Using Amazon EC2 (Ubuntu Server).

Hosting a Static Website Using Amazon EC2 (Ubuntu Server)

Prerequisites

- AWS Account
- Ubuntu EC2 Instance (Ubuntu 22.04 LTS or later)
- Security Group allowing:
 - SSH (Port 22)
 - HTTP (Port 80)
 - HTTPS (Port 443) (Optional)
- Key Pair (.pem file)
- Static website files (index.html, style.css, images, etc.)

Step 1: Launch an Ubuntu EC2 Instance

1. Login to AWS Management Console.
2. Navigate to EC2 Dashboard.
3. Click Launch Instance.
4. Select Ubuntu Server 22.04 LTS.
5. Choose t2.micro (Free Tier eligible).
6. Create or select a Key Pair.
7. Configure Security Group:
 - SSH → Port 22
 - HTTP → Port 80
 - HTTPS → Port 443 (optional)
8. Launch the instance.

Step 2: Connect to the EC2 Instance

From Linux/macOS

```
chmod 400 key.pem
```

```
ssh -i key.pem ubuntu@<Public-IP>
```

From Windows (PowerShell)

ssh -i key.pem ubuntu@<Public-IP>

Example

ssh -i mykey.pem ubuntu@13.234.xxx.xxx

Step 3: Update the Ubuntu Server

sudo apt update

sudo apt upgrade -y

Step 4: Install Apache Web Server

sudo apt install apache2 -y

Step 5: Start Apache

sudo systemctl start apache2

Enable Apache at boot

sudo systemctl enable apache2

Check status

sudo systemctl status apache2

Expected output

Active: active (running)

Step 6: Open the Default Web Page

Open the browser

http://Public-IP

Example

http://13.234.xxx.xxx

You should see

Apache2 Ubuntu Default Page

Step 7: Locate the Website Directory

Apache serves files from

`/var/www/html`

Check files

`cd /var/www/html`

ls

You will see

`index.html`

Step 8: Remove the Default Web Page

`sudo rm index.html`

Step 9: Upload Your Website

Method 1: Using SCP

From your local computer

`scp -i key.pem -r website/* ubuntu@Public-IP:/tmp`

Move files

`sudo cp -r /tmp/* /var/www/html/`

Method 2: Using Git

`sudo apt install git -y`

`cd /var/www/html`

`sudo git clone https://github.com/username/project.git .`

Method 3: Create Files Directly

`sudo nano /var/www/html/index.html`

Example

`<!DOCTYPE html>`

`<html>`

```
<head>
```

```
<title>My AWS Website</title>
```

```
</head>
```

```
<body>
```

```
<h1>Hello AWS!</h1>
```

```
<p>Website hosted on Amazon EC2.</p>
```

```
</body>
```

```
</html>
```

Save

Ctrl + O

Enter

Ctrl + X

Step 10: Set Proper Permissions

```
sudo chown -R www-data:www-data /var/www/html
```

```
sudo chmod -R 755 /var/www/html
```

Step 11: Restart Apache

```
sudo systemctl restart apache2
```

Step 12: Test the Website

Visit

<http://Public-IP>

Your website should appear.

Apache Commands

Start Apache

```
sudo systemctl start apache2
```

Stop Apache

```
sudo systemctl stop apache2
```

Restart Apache

```
sudo systemctl restart apache2
```

Reload Configuration

```
sudo systemctl reload apache2
```

Check Status

```
sudo systemctl status apache2
```

Website Directory

```
/var/www/html
```

Security Group Settings

Type	Port	Purpose
SSH	22	Remote Login
HTTP	80	Website Access
HTTPS	443	Secure Website

Directory Structure Example

```
/var/www/html
```

```
|
|— index.html
|— about.html
|— contact.html
|— css
|  └— style.css
```

```

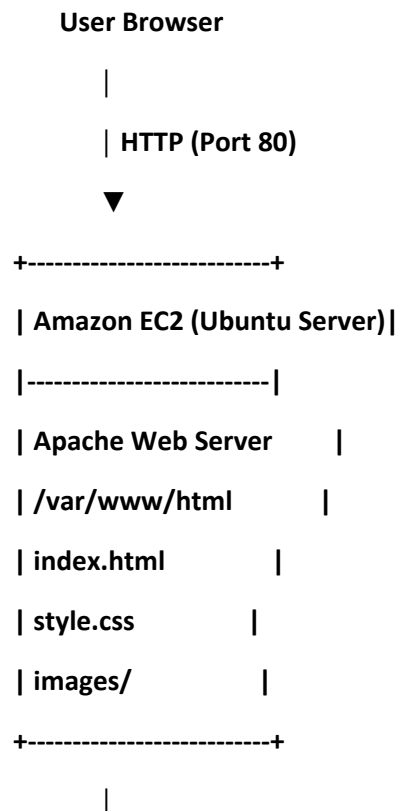
├─ js
|   └─ script.js
└─ images
    ├── logo.png
    └─ banner.jpg

```

Troubleshooting

Problem	Solution
Website not opening	Verify HTTP (Port 80) is allowed in the Security Group.
Permission denied	Run <code>sudo chown -R www-data:www-data /var/www/html</code> .
Apache not running	Execute <code>sudo systemctl restart apache2</code> .
Default Apache page still appears	Remove <code>/var/www/html/index.html</code> before copying your website files.
CSS/Images not loading	Ensure the folder structure and file paths match those referenced in your HTML.

Architecture





Static Website

Learning Outcomes

After completing this exercise, you will be able to:

- **Launch and configure an Ubuntu EC2 instance.**
- **Connect to the instance using SSH.**
- **Install and manage the Apache web server.**
- **Deploy a static website to `/var/www/html`.**
- **Configure file permissions and verify website accessibility.**
- **Access the hosted website using the EC2 instance's public IP address.**